MA.6.NSO.1.3

Benchmark

MA.6.NSO.1.3	**Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.**

Benchmark Clarifications:
Clarification 1: Instruction includes the connection of absolute value to mirror images about zero and to opposites.
Clarification 2: Instruction includes vertical and horizontal number lines and context referring to distances, temperature and finances.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.4, MA.6.NSO.4.1, MA.6.NSO.4.2
- MA.6.AR.1.1, MA.6.AR.1.2
- MA.6.GR.1.1, MA.6.GR.1.2, MA.6.GR.1.3

Terms from the K-12 Glossary

- Absolute Value
- Integers
- Number Line
- Rational Number
- Whole Number

Vertical Alignment

Previous Benchmarks
- This is the first introduction to the concept of absolute value.

Next Benchmarks
- MA.7.NSO.2.1

Purpose and Instructional Strategies

In elementary grades, students plotted positive numbers on a number line and related the addition of positive numbers to distance on a number line. In grade 6, students determine the absolute values, both in the context of the distance from zero and determining the distance between two points on the coordinate plane with the same x- or y-coordinate. In grade 7, students will use the concept of opposites when solving problems involving order of operations and absolute value.

- Instruction includes developing the understanding that absolute value explains the magnitude of a real number without regard to its sign and is denoted by $|x|$ and reads "the absolute value of x."
- Absolute value in real-life situations can help students understand the concept of absolute value.
  - Distance is not depicted as a negative number; thus, absolute value context should be used to describe a distance in the opposite direction. Students can draw pictures or diagrams, including vertical or horizontal number lines, to mathematically demonstrate what is happening in a real-life situation *(MTR.2.1, MTR.7.1)*.
- Using sentence frames can help students describe relationships and reasoning with absolute value *(MTR.4.1)*.
  - For example, when given the statement $|x| = 6$, students may benefit from a frame such

as: The distance from x to 0 is _____, so x can be located at _____ or _____.

- Instruction includes the use of technology to explore, interpret and define absolute value.

Common Misconceptions or Errors

Students may incorrectly state the absolute value of a negative number has a negative value. Students need to understand the total distance you traveled is not dependent on which direction you travel. To help address this misconception, instruction includes students talking about absolute value as distance and asking students questions such as:

- Would you describe how far you traveled with a negative number?
- If your parent drives a car in reverse, does the odometer show how far the car traveled by counting backwards?
- If you walk from your desk to the door backwards (your back is facing the door), did you walk a negative distance?
- You stand in line for a ride at an amusement park. You walk forward 10 feet in line, then the line makes a U-turn and you walk 30 feet. A U-turn happens in the line again and you travel an additional 16 feet before boarding the ride. How do you determine the distance you traveled while traveling in line?

Strategies to Support Tiered Instruction

- Instruction includes providing students with a sentence stem to interpret the meaning of the absolute value. Some students may require additional reading support.
    - For example, if given "If the temperature in Chicago, IL, is −7°, how many degrees below zero is the temperature?" the teacher can provide the sentence stem: "The absolute value of −7 is 7 units from zero, so the temperature is 7 degrees below zero."
- Instruction includes providing error analysis problems for which the absolute value of a positive number is incorrectly given as its opposite, rather than its distance from zero. Teacher reinforces the absolute value as the distance from zero and provides opportunities for students to plot the value on a number line and record the number of units the point is from zero. Instruction begins with integers and moves toward rational numbers.
- Teacher co-creates a graphic organizer with the students while providing instruction on the definitions of "absolute value," "opposite value," and "negative number." Instruction includes helping students to develop a definition in their own words, and identify key characteristics, examples, and non-examples of each term.
- Teacher provides students with flashcards to practice and reinforce academic vocabulary.
- Instruction includes students talking about absolute value as distance and asking students questions such as:
    - If your parent drives a car backwards, does the odometer show how far the car traveled by counting backwards?
    - If you walk from your desk to the door backwards (your back is facing the door), how far would you walk?
    - You stand in line for a ride at an amusement park. You walk forward 10 feet in line, then the line makes a U-turn and you walk 30 feet. A U-turn happens in the line again and you travel an additional 16 feet before boarding the ride. Do you subtract the distance when you travel the opposite direction in line, or do you still add it because you are still traveling over a specific distance?

- Do you describe how far you traveled with a negative number because a person was walking or driving backwards?

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.5.1)

The absolute value of an unknown number is 11.2. Where could the unknown number be located on a number line? Explain how you know.

Instructional Task 2 (MTR.7.1)

The table below shows the change in rainfall from the last 5 years. Find the absolute value of each month and determine which month had the greatest change in rainfall.

Month	March	April	May	June	July
Change in Rainfall Amount from 5-Year Average (inches)	0.21	-1.64	-0.48	2.01	-2.30

Instructional Task 3 (MTR.3.1)

Plot $4, -4$ and 0 on the same number line.

Part A. Compare the distance of 4 and -4 in relation to 0.

Part B. Explain how this relates to the definition of absolute value.

Instructional Items

Instructional Item 1

What is the value of the expression $\left|-\frac{7}{8}\right|$?

Instructional Item 2

If the temperature in Chicago, IL is $-7°$, how many degrees below zero is the temperature?

Instructional Item 3

What is the value of the expression $-|12.75|$?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered*